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RESEARCH ARTICLE

Impact of pharmaceutical care on knowledge, quality of life and satisfaction of postmenopausal women with osteoporosis

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Abstract *Background* This study describes the analysis of secondary outcomes from a previously published randomised controlled trial, which assessed the effects of pharmaceutical care on medication adherence, persistence and bone turnover markers. The main focus of this manuscript is the effect of the provision of pharmaceutical care on these secondary outcomes, and details on the design of the intervention provided, the osteoporosis care plan and materials used to deliver the intervention. *Objectives* To evaluate the effects of pharmaceutical care on knowledge, quality of life (QOL) and satisfaction of postmenopausal osteoporotic women prescribed bisphosphonates, and their associating factors. *Setting* Randomised controlled trial, performed at an osteoporosis clinic of a tertiary hospital in Malaysia. *Methods* Postmenopausal women diagnosed with osteoporosis (T-score ≤ -2.5/low-trauma fracture), just been prescribed weekly alendronate/risedronate were randomly allocated to receive intervention or standard care (controls). Intervention participants received a medication review, education on osteoporosis, risk factors, lifestyle modifications, goals of therapy, side effects and the importance of medication adherence at months 0, 3, 6 and 12. *Main outcomes measure* Knowledge, QOL and satisfaction. *Results* A total of 198 postmenopausal osteoporotic women were recruited: intervention = 100 and control = 98. Intervention participants reported significantly higher knowledge scores at months 3 (72.50 vs. 62.50 %), 6 (75.00 vs. 65.00 %) and 12 (78.75 vs. 68.75 %) compared to control participants. QOL scores were also lower (which indicates better QOL) at months 3 (29.33 vs. 38.41), 6 (27.50 vs. 36.56) and 12 (27.53 vs. 37.56) compared to control participants. Similarly, satisfaction score was higher in intervention participants (93.67 vs. 84.83 %). More educated women, with back pain, who were provided pharmaceutical care had better knowledge levels. Similarly, older, more educated women, with previous falls and back pain tend to have poorer QOL, whilst women who exercised more frequently and were provided pharmaceutical care had better QOL. Satisfaction also increased as QOL increases and when provided pharmaceutical care. *Conclusion* The provision of pharmaceutical care improved knowledge, QOL and satisfaction in Malaysian postmenopausal osteoporotic women, showing that pharmacists have the potential to improve patients' overall bone health. Policymakers should consider placing a clinical pharmacist in the osteoporosis clinic to provide counselling to improve these outcomes.

Keywords Knowledge · Malaysia · Osteoporosis · Patient satisfaction · Pharmaceutical care · Quality of life

Impact on Practice

- The provision of pharmaceutical care improved knowledge, quality of life (QOL) and satisfaction in Malaysian postmenopausal osteoporotic women

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- Pharmacists should consider providing pharmaceutical care to improve the overall bone health of patients
- The pharmaceutical care model derived from this study can also be applied to community pharmacists who are the first point of contact for many patients

Introduction

Osteoporosis is a major public health concern worldwide [1]. Although there is limited data on the prevalence of osteoporosis in Asia, it has been estimated that by the year 2050, 50 % of hip fractures worldwide will occur in Asia due to an increase in the elderly population [2]. If no prevention strategies are implemented, the incidence and costs related to osteoporotic fractures are expected to increase by 50 % over the next two decades [3].

Education regarding osteoporosis prevention seemed to encourage women to make lifestyle changes [4]. A systematic review [5] of interventions by healthcare professionals on community-dwelling postmenopausal women with osteoporosis reported that 2 out of 6 (33.3 %) studies showed an improvement in knowledge [6, 7], while others had no significant improvement [4, 8–10]. Intervention studies with positive impact involved more intensive education programmes such as individualised counselling, group-based behavioural education or the provision of an information leaflet on osteoporosis to reinforce verbal communications [6, 7]. Studies which did not show any effect provided information using passive methods such as the postal mail [10] or the telephone [9, 10]. The education methods used in these studies were also not individualized to patients' needs [4, 8].

In the same review [5], 8 out of 13 (61.5 %) studies showed a significant improvement in QOL of the intervention group [11–18], whilst another five reported no statistical difference [9, 19–22]. This may be due to the low intensity of intervention, the high QOL at baseline, lack of adherence to intervention or the short study duration of 6 months [5]. In addition, studies which showed a significant improvement in the intervention group had appropriate control groups which were given only usual pharmacy services [5].

Randomised controlled trials to assess patient satisfaction are scarce. Only one study reported an improvement in satisfaction scores [23] whilst another two found no difference [9, 24]. This may be attributed to the low intensity of intervention provided or the use of generic satisfaction instruments [23, 24].

To date, only prospective, observational studies have been conducted to assess the effectiveness of pharmacist interventions in the management of osteoporosis; and not randomised control trials. The provision of pharmaceutical care showed that screening and early detection [25–27], monitoring of drug therapy [28], provision of patient education [25–28] and follow-up counselling [28] resulted in positive patient outcomes. However, in Malaysia, there is no documented evidence on the involvement of pharmacists in the management of osteoporosis.

Aim of the study

To evaluate the effects of pharmaceutical care on knowledge, QOL and satisfaction of postmenopausal women with osteoporosis.

Methods

This study describes the analysis of secondary outcomes from a previously published randomised controlled trial, which assessed the effects of pharmaceutical care on medication adherence, persistence and bone turnover markers [29, 30].

Since once weekly bisphosphonates were the most widely prescribed medication for osteoporosis in the hospital under study, patients who were either on the once weekly alendronate [Fosamax®, Merck Sharp & Dohme (Italia) S.P.A., Pavia, Italy] or risedronate (Actonel®, OSG Norwich Pharmaceuticals, Incorporation, North Norwich, NY, USA) were recruited in this study. Participants were randomly allocated to the control and intervention group as described previously [29].

Participants

Postmenopausal women diagnosed with osteoporosis with a T-score ≤ -2.5 at any bone site or with a previous low-trauma fracture, and had just been prescribed alendronate or risedronate were recruited. Inclusion criteria were participants who have never been on any active osteoporosis therapy within the past 6 months, and were able to communicate in English. Excluded were those with metabolic bone or bone metabolism disease, a history of chronic renal, hepatic or gastrointestinal disease or traumatic lumbar compression fracture. Pharmaceutical care was provided to the intervention group while the control group received standard pharmacy service in the hospital.

This study was approved by the hospital's Medical Ethics Committee (ethics approval no: 442.7) before commencement of the study. All participants provided their informed written consent.

Sample size

The sample size was calculated based on the primary outcome as mentioned previously [29]. However, it was also postulated that there would be a mean difference of 5 % in knowledge, QOL and satisfaction between the control and intervention group, with a pooled standard deviation of 10 %. To obtain a 80 % power of detection and $\alpha = 0.05$, a sample size of at least 64 was required in each group [31]. Assuming a 20 % loss to follow-up, the total number of participants required was at least 77 in each arm.

Intervention and main outcome measures

This study involved four visits over a 1-year period. Pharmaceutical care was provided by a practicing hospital pharmacist with a basic degree in pharmacy and 10 years of clinical experience. Only one pharmacist was involved in the provision of pharmaceutical care in the present study to ensure consistency of information given. An osteoporosis care sheet was developed to guide the pharmacist during the counselling sessions and to record all findings. This pharmacist was then trained on how to use the osteoporosis care sheet. A pilot study was then conducted.

Participants in the intervention group received a "pharmaceutical care package" which included a one-to-one, individualized medication review, education on osteoporosis, risk factors, lifestyle modifications, goals of therapy, side effects and the importance of adherence, at months 0 (baseline), 3, 6 and 12. Materials used included a booklet on osteoporosis and a personalized medication regimen which were given to intervention participants. The pharmacist spent between 20 and 40 min per intervention participant. Medication review was limited to osteoporosis medications to maintain the focus of the study objectives. Monthly follow-ups via telephone calls were conducted for the first 6 months, then every 3 months up to month 12.

Control participants received standard pharmacy service where the pharmacist dispensed the osteoporosis medications with only explanation of how to take the medications (after getting up for the day, in an upright position with a full glass of water and 30 min before breakfast).

At baseline, demographic data was collected and bisphosphonate therapy was initiated. Knowledge and QOL of participants were assessed at months 3, 6 and 12, whilst satisfaction was assessed at month 6 only.

Instruments used

The validated English version of the Malaysian Osteoporosis Knowledge Tool (MOKT) [32], the Quality of Life Questionnaire of the European Foundation for Osteoporosis (QUALEFFO) [33] and the Osteoporosis Patient Satisfaction Questionnaire (OPSQ) [34] were used to measure the patient's knowledge, QOL and satisfaction, respectively.

Statistical analyses

All data was analysed using the Statistical Package for Social Sciences (SPSS) version 15 (Chicago, IL, USA). Continuous data were expressed as mean ± standard deviation (SD). Categorical variables were expressed as absolute (number) and relative frequencies (percentage). The demographic data of participants were compared using the independent t test for continuous variables while Pearson's χ^2 test was used for comparison of proportions between the two groups.

The effects of pharmaceutical care on QOL, knowledge and patient satisfaction was analysed using non-parametric Mann–Whitney U test as normality assumptions could not be fulfilled. To compare results within each group over time, the Friedman test (since there were three visits in total) and the Wilcoxon-Signed Rank test (for comparison between two visits) were used. A multiple linear regression model was then performed to determine the association between these factors with knowledge, QOL and satisfaction. A p value <0.05 was considered as statistically significant.

Results

A total of 198 patients were recruited (intervention = 100, control = 98). The flow of participants throughout the study is shown in Fig. 1. No significant difference was found between the control and intervention group in all demographic aspects, as reported previously [29].

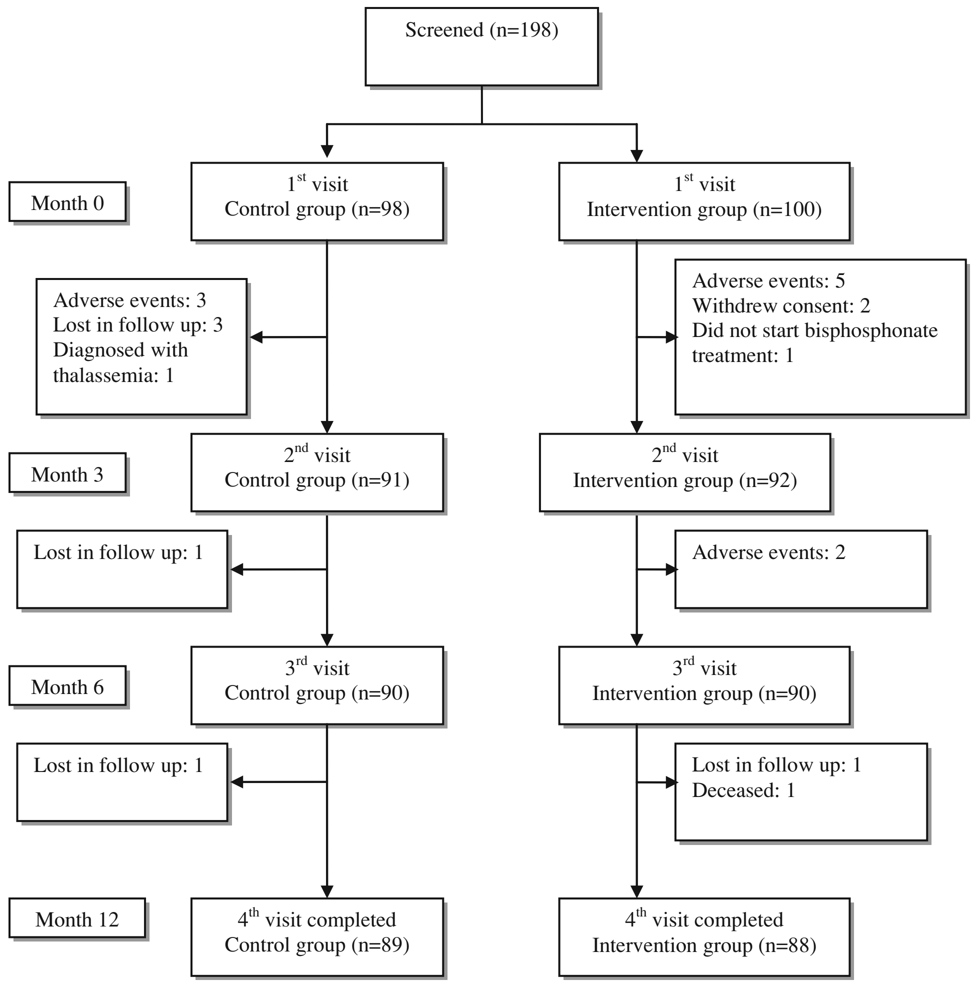
Knowledge

The internal consistency of the overall MOKT in the present study is excellent (Cronbach's $\alpha = 0.90$) and is similar to that of the MOKT validation study [32].

Knowledge scores were higher at months 3 (72.50 vs. 62.50 %), 6 (75.00 vs. 65.00 %) and 12 (78.75 vs. 68.75 %) compared to control participants, indicating the impact of individualised pharmaceutical care on knowledge (Table 1). Both control & intervention participants showed significant improvement in knowledge scores over time. There was a significant increase in total knowledge scores in the control group from month 3 to 6, as well as from month 3 to 12 (both with $p < 0.001$). For the intervention group, the total knowledge scores increased significantly from month 3 to 6 as well as from month 3 and 6 to month 12 (all $p < 0.001$).

The linear combination of education level, back pain and whether the participant was provided pharmaceutical care or not, was significantly related to knowledge levels, $F(3, 179) = 26.18$, $p < 0.01$. The sample multiple

Fig. 1 Summary of number of participants in the study



correlation coefficient was 0.55, indicating that approximately 30.5 % of the variance of knowledge levels can be accounted for by the linear combination of these factors: knowledge level = 38.98 + 7.299 (education level) + 7.604 (back pain) + 11.000 (pharmaceutical care). Women who were more educated, with back pain and who were provided pharmaceutical care, had better knowledge levels. No collinearity was found in this multiple regression analysis.

Quality of life (QOL)

The overall internal consistency of the QUALEFFO in the present study is high (Cronbach's $\alpha = 0.88 - 0.92$) and this is similar to that obtained in the QUALEFFO validation study [33].

Intervention participants had significantly better QOL scores (i.e. a lower score represents better QOL) at months 3 (29.33 vs. 38.41), 6 (27.50 vs. 36.56) and 12 (27.53 vs. 37.56) compared to control participants, indicating the positive impact of pharmaceutical care (Table 2). There was no significant change over time within the control and intervention group at months 3, 6 and 12. The only exception was the general health perception domain in the intervention group where significant difference in QOL scores was observed between month 3 and 6 ($p = 0.004$).

The linear combination of age, education level, previous falls, back pain, frequency of exercise and whether the participant was provided pharmaceutical care or not, was significantly related to QOL scores, $F(6, 176) = 14.77$, $p < 0.01$. The sample multiple correlation coefficient was 0.58, indicating that approximately 33.5 % of the variance of QOL can be accounted for by the linear combination of these factors: QOL score = 9.76 + 0.456 (age) − 3.018 (level of education) + 5.172 (previous falls) + 3.795 (back pain) − 4.047 (frequency of exercise) − 5.858 (pharmaceutical care). Women who were older, more educated, with previous falls and back pain tended to have poorer QOL; whereas women who exercised more frequently and were provided pharmaceutical care had better QOL. No collinearity was found in this multiple regression analysis.

Table 1 Knowledge scores of the control and intervention group

	Month 3								
	Control (n = 91)			Intervention (n = 92)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
General information	60.66	32.59	60.00	76.52	22.89	80.00	79.59 104.30	−3.251	0.001*
Consequences	55.05	21.62	60.00	65.33	13.05	70.00	79.16 104.70	−3.339	0.001*
Risk factors	44.76	24.04	46.67	57.17	17.82	60.00	78.21 105.64	−3.520	<0.001*
Treatment	80.66	18.49	90.00	89.46	12.52	90.00	77.29 106.55	−3.866	<0.001*
Total knowledge score	58.30	19.01	62.50	69.70	11.27	72.50	75.81 108.02	−4.122	<0.001*
	Month 6								
	Control (n = 90)			Intervention (n = 90)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
General information	68.89	26.83	80.00	82.00	20.12	80.00	77.52 103.48	−3.488	<0.001*
Consequences	64.00	15.71	70.00	69.11	14.66	70.00	83.25 97.75	−1.927	0.054
Risk factors	52.44	21.32	53.33	64.07	17.62	66.67	75.97 105.03	−3.762	<0.001*
Treatment	83.33	16.76	90.00	91.11	11.16	90.00	77.02 103.98	−3.624	<0.001*
Total knowledge score	65.11	15.01	65.00	74.33	11.33	75.00	72.77 108.23	−4.574	<0.001*
	Month 12								
	Control (n = 88)			Intervention (n = 88)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
General information	71.59	26.04	80.00	84.77	21.65	100.00	74.49 102.51	−3.848	<0.001*
Consequences	62.16	18.78	70.00	73.18	13.18	70.00	75.09 101.91	−3.604	<0.001*
Risk factors	52.05	22.64	53.33	67.42	19.19	73.33	71.01 105.99	−4.580	<0.001*
Treatment	84.77	18.94	90.00	93.86	7.65	100.00	74.97 102.03	−3.753	<0.001*
Total knowledge score	65.20	17.24	68.75	77.64	11.70	78.75	68.05 108.95	−5.337	<0.001*

* Statistically significant at $p < 0.05$

SD standard deviation

Table 2 Quality of life scores in the control and intervention group

	Month 3								
	Controls (n = 91)			Intervention (n = 92)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
Pain	2.29	0.97	2.40	1.90	0.87	1.80	103.12 81.01	−2.850	0.004*
Physical function	2.14	0.77	1.94	1.79	0.53	1.71	103.30 80.82	−2.873	0.004*
Social function	3.36	1.00	3.43	2.83	0.89	2.74	106.74 77.42	−3.743	<0.001*
General health perception	3.39	0.88	3.33	3.12	0.83	3.33	100.36 83.73	−2.140	0.032*
Mental function	2.53	0.59	2.56	2.43	0.52	2.33	98.92 85.16	−1.761	0.078
QUALEFFO score	38.05	14.76	38.41	30.15	11.64	29.33	106.28 77.88	−3.627	<0.001*
	Month 6								
	Controls (n = 90)			Intervention (n = 90)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
Pain	2.16	1.00	2.00	1.78	0.75	1.80	100.29 80.71	−2.560	0.010*
Physical function	2.11	0.78	2.00	1.78	0.53	1.71	100.92 80.08	−2.684	0.007*
Social function	3.26	1.04	3.46	2.76	0.86	2.69	104.34 76.66	−3.564	<0.001*
General health perception	3.30	0.91	3.33	2.91	0.76	3.00	102.52 78.48	−3.119	0.002*
Mental function	2.58	0.65	2.67	2.38	0.48	2.44	101.28 79.72	−2.783	0.005*
QUALEFFO score	37.01	16.40	36.56	28.22	10.98	27.50	104.74 76.26	−3.666	<0.001*
	Month 12								
	Controls (n = 88)			Intervention (n = 88)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
Pain	2.09	0.88	2.00	1.85	0.86	1.80	95.95 81.05	−1.961	0.050
Physical function	2.14	0.78	1.94	1.75	0.56	1.68	101.16 75.84	−3.300	0.001*
Social function	3.47	1.29	3.62	2.66	0.85	2.71	105.95 71.05	−4.544	<0.001*
General health perception	3.33	0.94	3.33	2.95	0.81	3.00	99.31 77.69	−2.834	0.005*
Mental function	2.65	0.68	2.72	2.35	0.56	2.33	101.30 75.70	−3.339	0.001*
QUALEFFO score	38.15	16.30	37.56	28.21	12.64	27.53	104.28 72.72	−4.108	<0.001*

* Statistically significant at $p < 0.05$

SD standard deviation

Table 3 Patient satisfaction of the control and intervention group

	Month 6								
	Control (n = 70)			Intervention (n = 72)			Mann–Whitney *U* test		
	Mean	SD	Median	Mean	SD	Median	Mean rank	z-value	*p* value
Clinical pharmacy service	87.27	7.72	88.00	92.60	7.22	95.00	72.04 108.96	−4.773	<0.001*
Usefulness of counselling	81.37	10.29	80.00	91.19	8.38	93.33	66.88 114.12	−6.123	<0.001*
Total satisfaction score	84.32	7.48	84.83	91.89	7.22	93.67	66.19 114.81	−6.262	<0.001*

* Statistically significant at $p < 0.05$

Patient satisfaction

The internal reliability of the overall OPSQ was excellent (Cronbach's $\alpha = 0.86$). Factor analysis of the OPSQ extracted one component, indicating that the OPSQ measures only one parameter (i.e. patient satisfaction). These results are similar to that obtained in the OPSQ validation study [35].

The overall OPSQ satisfaction score was higher in intervention participants ($p < 0.001$), indicating that participants who were provided with pharmaceutical care were more satisfied than those who received standard pharmacy care (Table 3). In addition, intervention participants also rated higher satisfaction levels than control participants in the clinical pharmacy service and usefulness of counselling provided.

The linear combination of QOL and whether the participant was provided pharmaceutical care or not, was significantly related to satisfaction levels, $F(2,\ 177) = 28.78$, $p < 0.01$. The sample multiple correlation coefficient was 0.50, indicating that approximately 24.5 % of the variance of satisfaction levels can be accounted for by the linear combination of these factors: satisfaction level = 88.36 − 0.109 (QOL) + 6.615 (pharmaceutical care). Satisfaction levels increases as QOL increases, and when provided pharmaceutical care. No collinearity was found in this multiple regression analysis.

Discussion

Intervention participants who received pharmaceutical care reported significantly higher knowledge, QOL and satisfaction than control participants, indicating the positive impact of the pharmaceutical care provided.

The knowledge level of intervention participants was significantly higher than control participants. This is probably due to the intensive pharmaceutical care sessions provided by the pharmacist. Higher education level and the presence of back pain were found to be positively correlated to the knowledge level of participants. A possible explanation is that women with back pain are more concerned about their health and hence, more motivated to seek knowledge concerning osteoporosis. In addition, a higher education level equipped them with the means to seek information. Generally, women in Asia have less knowledge about osteoporosis than women in developed countries. This may be due to a wider exposure to health issues and more established patient education programmes in developed countries. This indicates that pharmacists and other healthcare professionals in Malaysia can play a more active role in educating patients about bone health, especially in those whose knowledge and adherence to medication are poor.

The present study also showed that intervention participants had significantly better QOL in all domains as well as the overall QOL scores as compared to control participants. The only exception was the mental health domain at month 3; indicating that the pharmaceutical care provided was effective in improving functional activities, pain perception and overall QOL outcomes. Another reason for the improved QOL could be that medication adherence was better in the intervention group and this improved the pain scores, leading to an improved QOL assessment [36]. Previous studies also showed that an improvement in QOL was associated with an increase in age and knowledge [37] whereas, daily back pain was associated with reduced QOL, mobility, longevity and increased risk of coronary heart events [38].

Intervention participants who were provided pharmaceutical care showed significantly higher satisfaction levels than control participants. Ensuring that the medication regimen be kept as simple as possible, negotiating priorities with the patient, providing clear instructions, reminding patients about their appointments, reinforcing the importance of medication adherence at each visit and educating patients on osteoporosis are practical and effective services which could improve patient's satisfaction concerning health care services. Participants with better QOL were more likely to have higher patient satisfaction.

One of the limitations in this study is that data was collected only at one site and therefore, cannot be generalized to all Malaysian postmenopausal women with osteoporosis. However, the provision of pharmaceutical care to osteoporosis patients can be applied in any osteoporosis clinic (as per present study) or community pharmacy, with some basic training to familiarise the pharmacist with the osteoporosis care sheet. The resources required for a pharmacist who wishes to provide pharmaceutical care services would be their time and willingness to undergo training. The osteoporosis booklet and medication review regimen can be produced at minimal cost. Further studies should include pharmacoeconomic analysis of the services provided in order to convince stakeholders and policy makers on the importance of such services.

Conclusion

The provision of pharmaceutical care improved the knowledge, QOL and satisfaction in Malaysian postmenopausal osteoporotic women, showing that pharmacists have the potential to improve patients' overall bone health. Future pharmacoeconomic studies to determine the cost benefit of providing such pharmaceutical care should be performed to influence policy changes.

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Conflicts of interest None.

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